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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 310 (July 2023)
Communication Laboratory

Final Project Report

Section**: B1** Group**: 05**

Design Of Sampler, Quantizer And Encoder Scheme For Pulse Code Modulation (PCM)

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Academic Honesty Statement:

"In signing this statement, We, hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1 Abstract

A popular method for digitizing analog signals, pulse code modulation (PCM), finds utility in audio recording, telecommunications, and many other industries. The sampler, quantizer, and encoder – three basic parts that are necessary for PCM systems – are the main subjects of this project's design and execution. The continuous-time analog signal must be discretized by the sampler into a series of discrete amplitude values. The technique of transferring each sampled amplitude value into a limited number of discrete levels is known as quantization. The quantized amplitude values are transformed by the encoder into a binary representation that may be stored or transmitted. This project investigates design strategies for every part, taking into account variables like computational complexity, quantization levels, encoding methods, and sampling rate. To assess the performance of the designed methods in terms of signal accuracy and efficiency, simulations and real-world implementations will be carried out.

2 Introduction

Pulse Code Modulation or PCM is a type of modulation where a message signal is represented by a sequence of coded pulses. This is achieved by encoding the signal in discrete form in terms of both time and amplitude. It is the foundation of contemporary communication and data storage systems and is widely used in telecommunications, audio recording, instrumentation, and other fields. At the heart of PCM lie three crucial components: the sampler, quantizer, and encoder. These factors impact the digitization process's overall integrity, effectiveness, and durability, thus requiring meticulous design and optimization.

The sampler starts the PCM process by discretizing the continuous-time analog signal into a sequence of discrete amplitude values, which is an important step in retaining the signal's key features. It is essential to use appropriate sampling techniques in order to minimize aliasing artifacts and precisely capture the analog waveform. The quantizer maps each sampled amplitude value onto a limited number of discrete levels after sampling. Within the constraints of finite digital precision, the quantizer attempts to accurately represent the analog signal by carefully balancing quantization noise and resolution. The encoder converts the quantized amplitude values into a binary format that is suitable for transmission or storage once they have been quantized. This is where the encoding scheme selection is crucial because it affects decoding difficulty, error resilience, and data bandwidth needs. To maximize data transmission speeds and maintain signal integrity in the face of noise and channel distortions, efficient encoding techniques are crucial.

This project embarks on a comprehensive exploration of the design and implementation of sampler, quantizer, and encoder schemes tailored for PCM systems. Drawing upon

established theories and methodologies in communication engineering, it aims to provide efficient and robust solutions capable of meeting the requirements of practical applications. Through theoretical analysis, simulation studies, and practical implementations, this endeavor seeks to advance our understanding of PCM systems while providing tangible contributions to signal digitization, transmission, and storage technologies. Ultimately, the insights gained from this project are poised to inspire innovations reliant on PCM, ushering in new possibilities for communication and data processing.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of this project lies in the implementation of a useable circuitry for sampler, quantizer and encoder upon studying the system requirements. To be more specific, the goal of this project is to design a Pulse Code Modulation scheme, simulation of necessary circuits, hardware design, testing, prototyping and validation of the proposed design.

3.1.2 Literature Review

The design of samplers, quantizers, and encoder schemes is pivotal in shaping the performance of Pulse Code Modulation (PCM) systems. A thorough examination of the existing literature reveals key advancements and challenges in the field, laying the groundwork for the current project.

1. Sampler Design: Sampling, as a fundamental process in PCM, has been extensively studied in the literature. Classic sampling theorems, such as Nyquist-Shannon theorem, establish the theoretical foundations for sampling analog signals. Various sampling techniques, including uniform and non-uniform sampling, have been explored. However, the literature highlights challenges in achieving real-time performance and accurate representation of high-frequency signals.

2. Quantization Techniques: Quantization, the process of mapping continuous amplitude values to discrete levels, plays a crucial role in determining the fidelity of the digitized signal. Literature in quantization encompasses traditional uniform quantization methods and more recent approaches such as non-uniform quantization. Existing research indicates that quantization noise remains a significant challenge, particularly in low-bit systems. Advanced quantization algorithms, including oversampling and noise shaping, aim to mitigate these issues. The project's focus on optimizing the quantization process aligns with the identified need for reducing quantization noise and maintaining signal fidelity.

3. Encoding Schemes for PCM: Efficient encoding is essential for minimizing data transmission and storage requirements in PCM systems. Classic PCM encoding methods, such as Pulse Code Modulation and Delta Modulation, are well-established in the literature. However, advancements in compression techniques, such as Huffman coding and arithmetic coding, present opportunities for further optimization.

Literature emphasizes the importance of error detection and correction mechanisms in PCM encoding to ensure reliable data transmission. The current project incorporates these insights into the design process, aiming to develop an efficient encoding scheme with built-in error management.

4. Integration of Samplers, Quantizers, and Encoders: While individual components have been extensively studied, literature on the holistic integration of samplers, quantizers, and encoders for PCM systems is relatively limited. The need for a seamless and optimized interaction between these components is evident for achieving overall system efficiency and accuracy. This project bridges the existing gap by proposing a comprehensive design that considers the synergies between the sampler, quantizer, and encoder.

Conclusion: The literature review highlights the evolution of sampling, quantization, and encoding techniques in PCM systems. It identifies challenges and opportunities for improvement, providing a solid foundation for the current project's objectives. By building upon existing knowledge, this project seeks to contribute to the advancement of PCM systems, addressing key limitations and fostering innovation in the design of samplers, quantizers, and encoder schemes.

3.1.3 Formulation of Problem

The existing Sampler, Quantizer, and Encoder schemes in PCM systems exhibit certain limitations and challenges that hinder their optimal performance. These challenges include:

- Sampling Accuracy: The current sampling method may not capture the analog signal with the required precision, leading to inaccuracies in the representation of the signal.
- Quantization Noise: The quantization process introduces noise, affecting the fidelity of the reconstructed signal. Minimizing quantization error is crucial for maintaining signal quality.
- Encoding Efficiency: The encoding scheme may not be optimized for efficient data representation, potentially leading to increased data transmission requirements and storage demands.

Addressing these challenges is paramount for enhancing the overall performance of PCM systems. Improved sampling accuracy, reduced quantization noise, and an efficient encoding scheme will collectively contribute to a more reliable and high-fidelity representation of analog signals in digital form.

3.1.4 Analysis

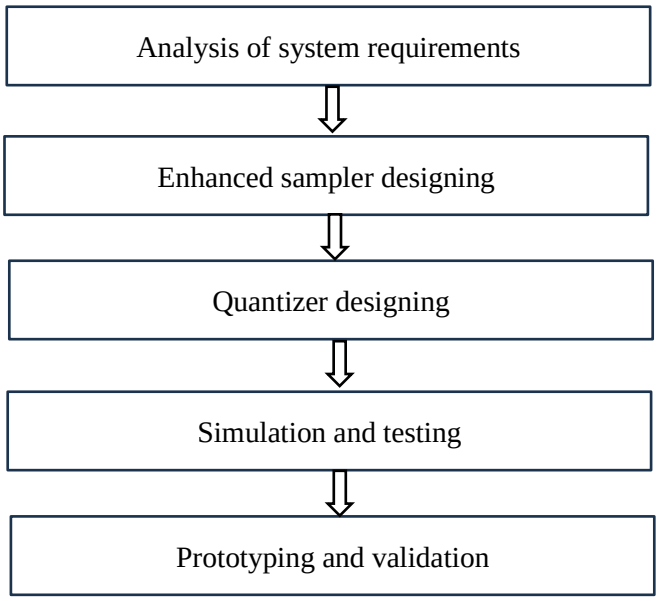
The analysis work done in this project are: individual evaluation of sampler, quantizer and encoder circuits, debugging and optimization of circuits, ensuring the use of most appropriate components and matching hardware output with simulation results.

3.2 Design Method (PO(a))

Before initiating the design process, a comprehensive analysis of the system requirements were conducted. This involved understanding the specifications and constraints of the PCM system, considering factors such as signal bandwidth, resolution, and desired fidelity. The aim was to establish a clear set of design parameters that will guide the development of the Sampler, Quantizer, and Encoder components.

Designing an enhanced sampler focused on increasing accuracy by considering factors such as signal frequency, Nyquist theorem compliance, and adaptive sampling techniques. The Quantizer was designed to minimize quantization noise while maintaining a suitable bit resolution.

Simulations and comparative analyses were performed to validate the effectiveness of the chosen algorithm. The simulations assessed the performance against predefined criteria, allowing for iterative refinement of the design. Upon successful simulation results, physical prototypes of the designed components were developed. These prototypes underwent practical testing and validation against real-world analog signals.



Block Diagram

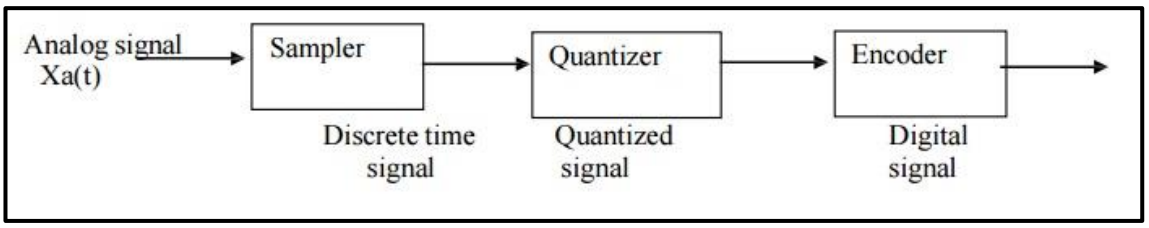


Fig: Block Diagram of A/D Conversion

3.3 Schematic Diagram

Below is the schematic of encoder, which we designed in PSpice. Here IC **74LS148** has been used as encoder. Moreover, MOSFETs and Op-Amps have been used.

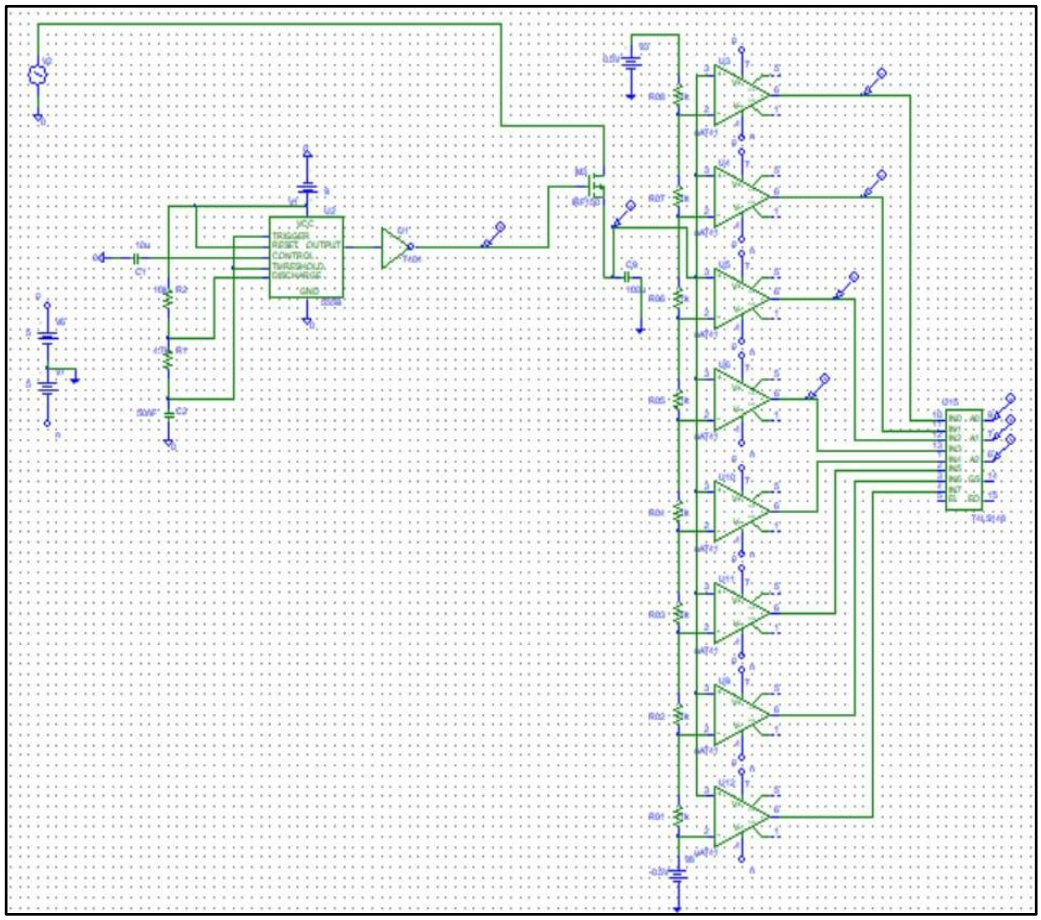
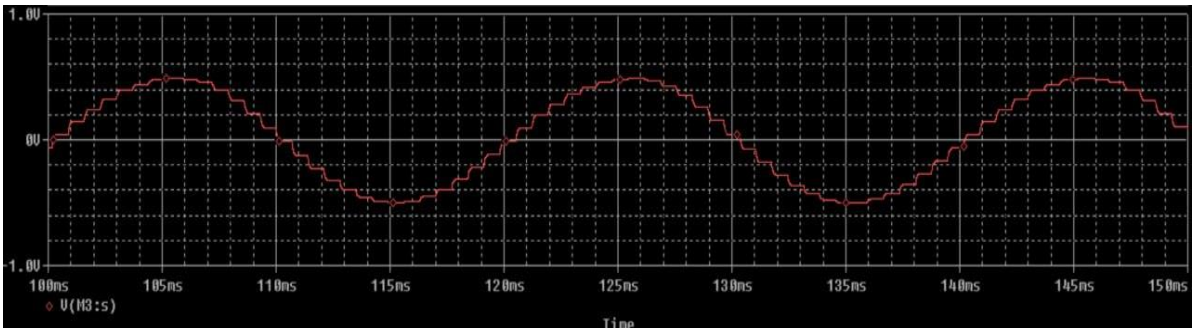


Fig: Schematic diagram of sampler, quantizer, and encoder

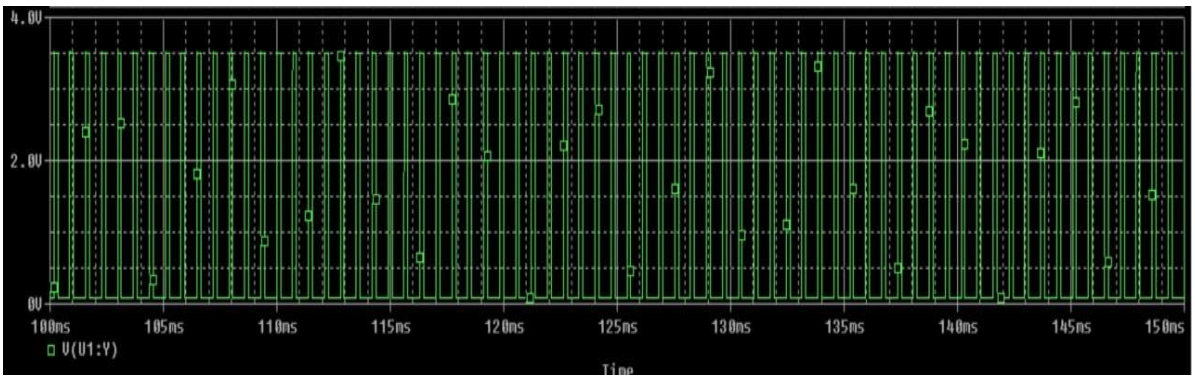
3.4 Schematic Outputs:

The outputs obtained from schematic are as follows:

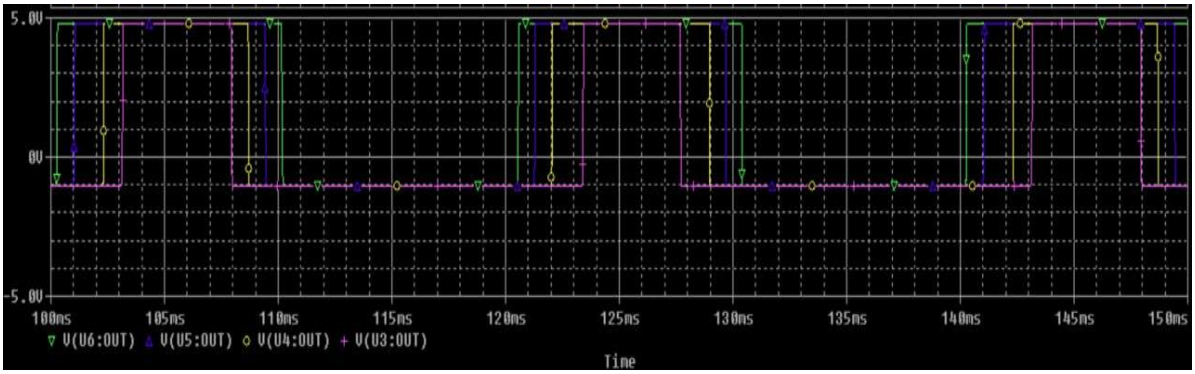
Sample and Hold Output:



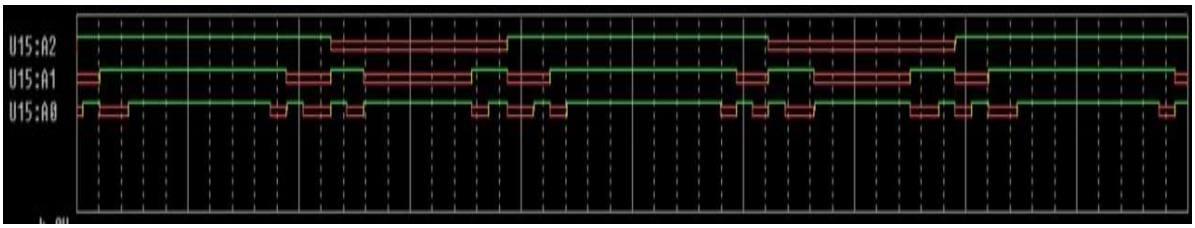
Pulse train:



Sequential Output After Quantization:



Output of Quantizer:



4 Implementation

4.1 Description

We designed the following encoder in Proteus software using logic gates, and it was later implemented as hardware setup:

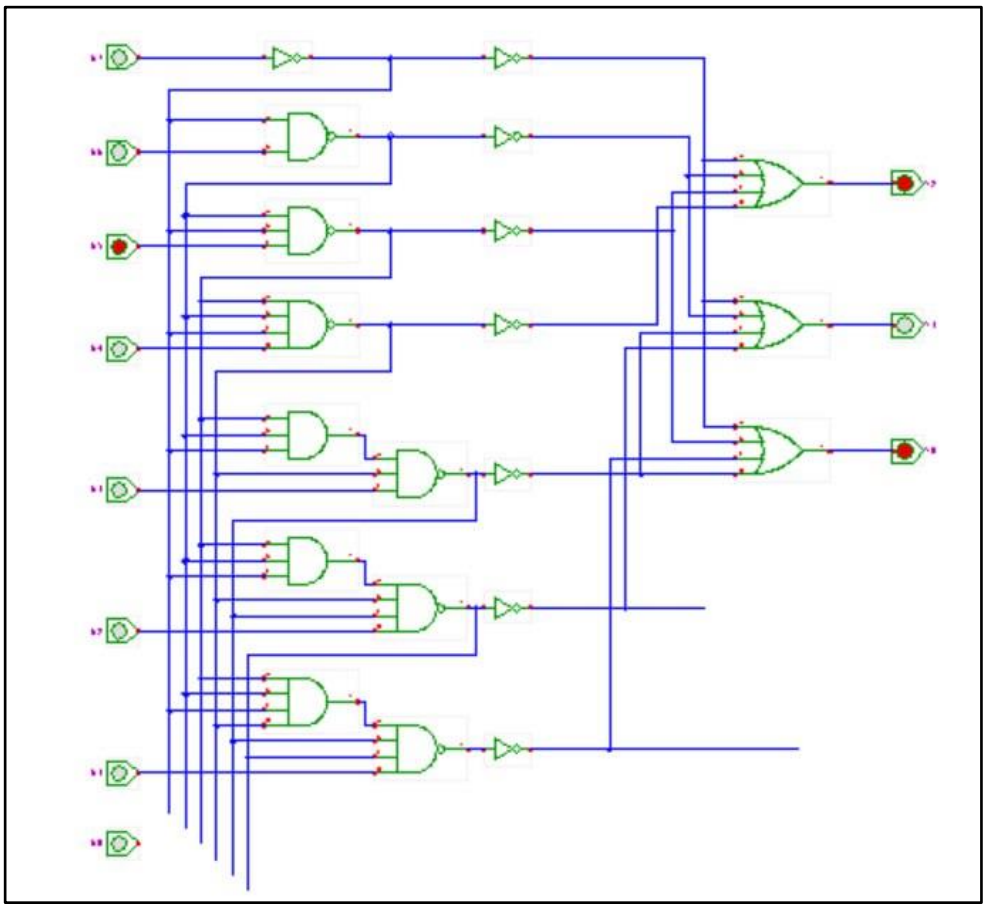
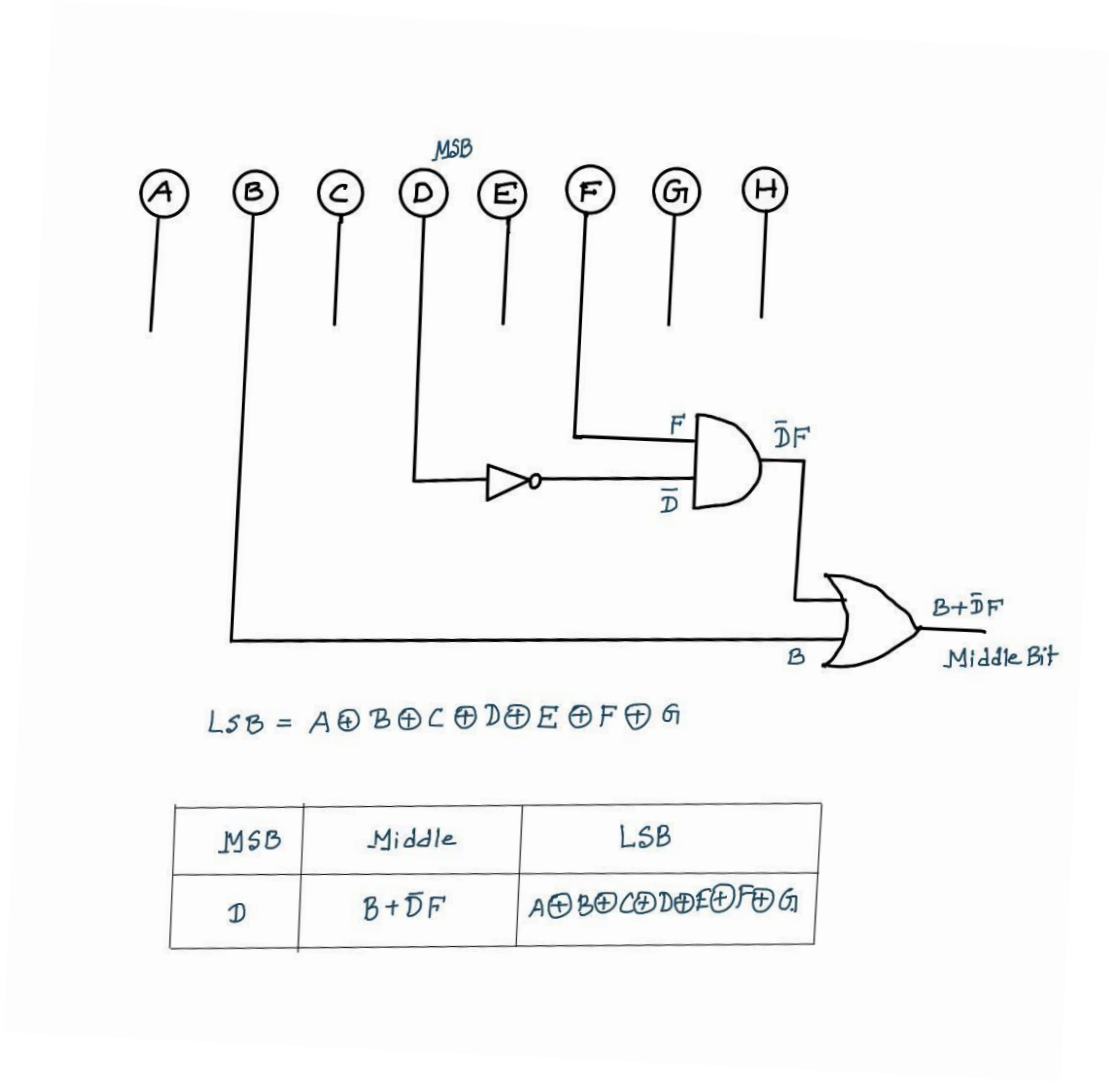


Fig: Logic Diagram of Encoder in Proteus

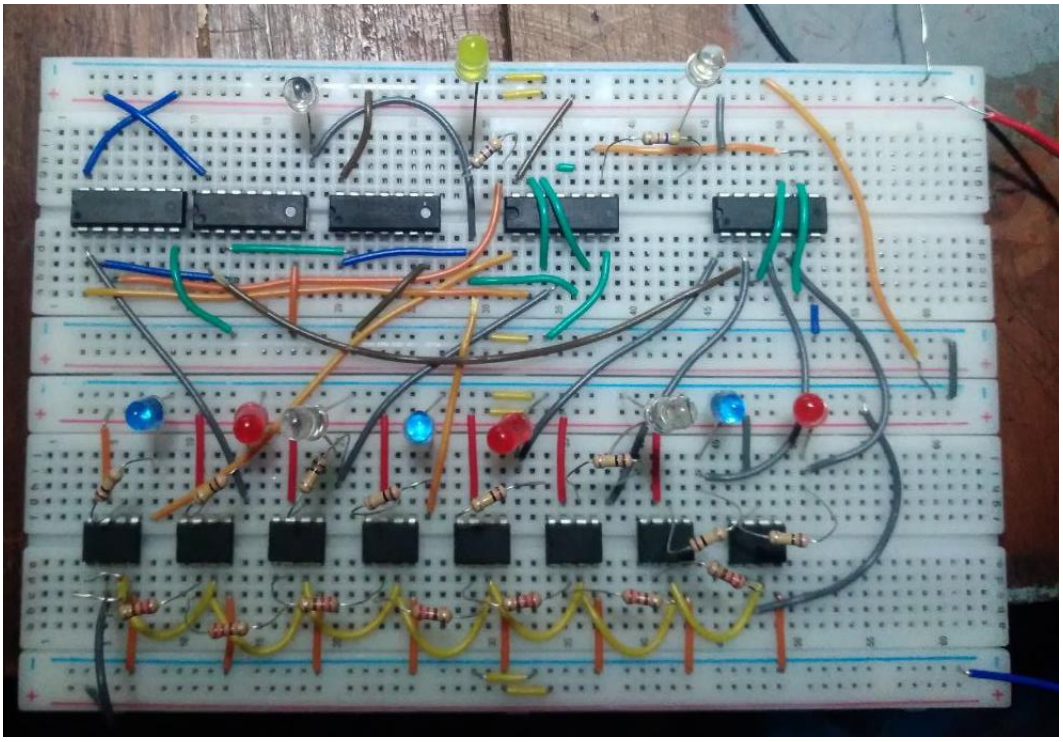
Here, we have explained the inputs and outputs, the MSB (Most Significant Bit), middle bit, and the least significant bit of the outputs:



MSB	Middle	LSB
D	$B+\bar{D}F$	A⊕B⊕C⊕D⊕E⊕F⊕G

4.2 Hardware Implementation

This is how we designed the circuit. Basically we implemented the software design shown previously.



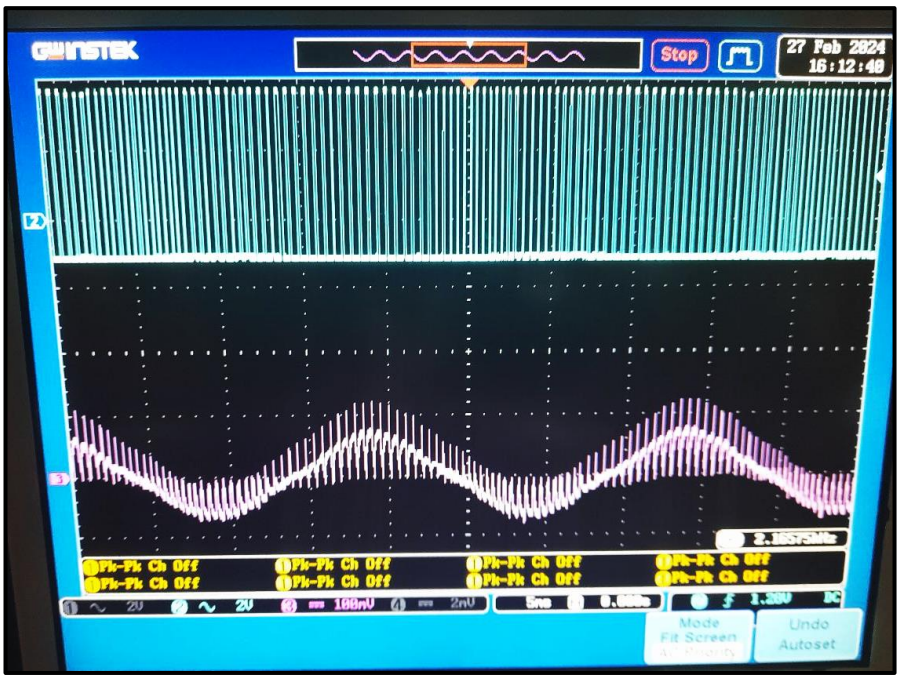
4.3 Experiment and Data Collection

We formed the following truth table using the inputs and outputs from the encoder we used in our project.

								MSB	Mid	LSB
A	B	C	D	E	F	G	H	I1	I2	I3
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	1	0	0	1
0	0	0	0	0	1	1	1	0	1	0
0	0	0	0	1	1	1	1	0	1	1
0	0	0	1	1	1	1	1	1	0	0
0	0	1	1	1	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1

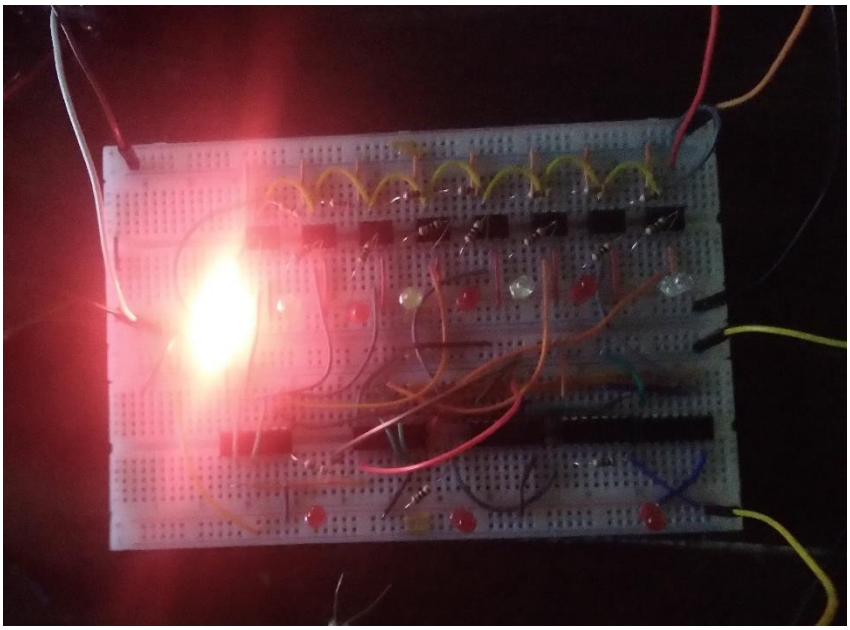
4.4 Results

Sampler Output: This output was observed using oscilloscope.

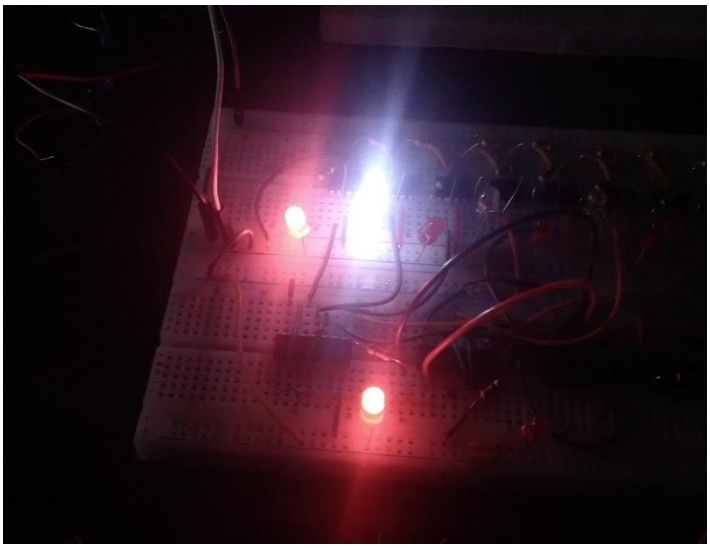


Encoder Output:

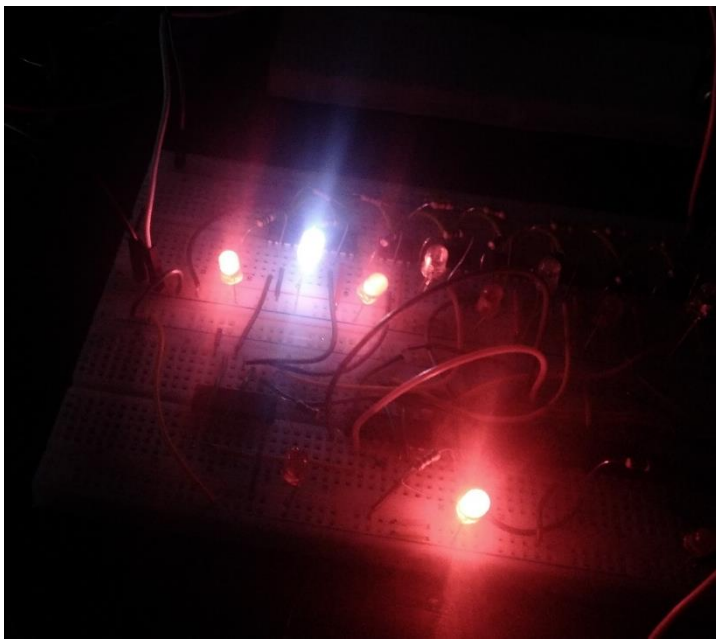
Case 1: When only one input is 1, and no output LEDs are on:



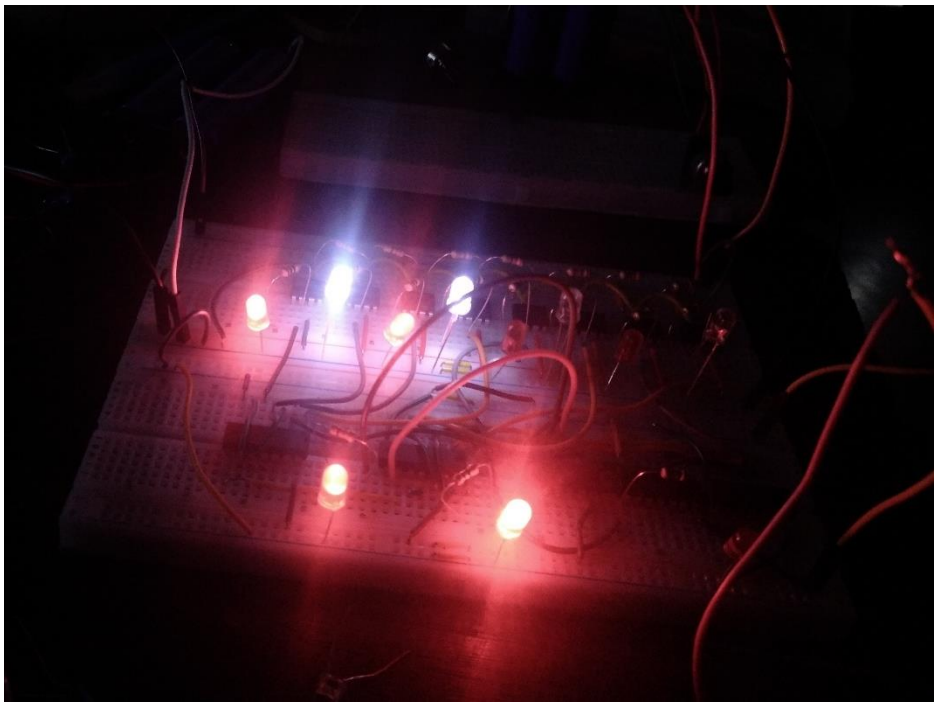
Case 2: When only two inputs are 1, and only the 3rd output LED(I3) is on:



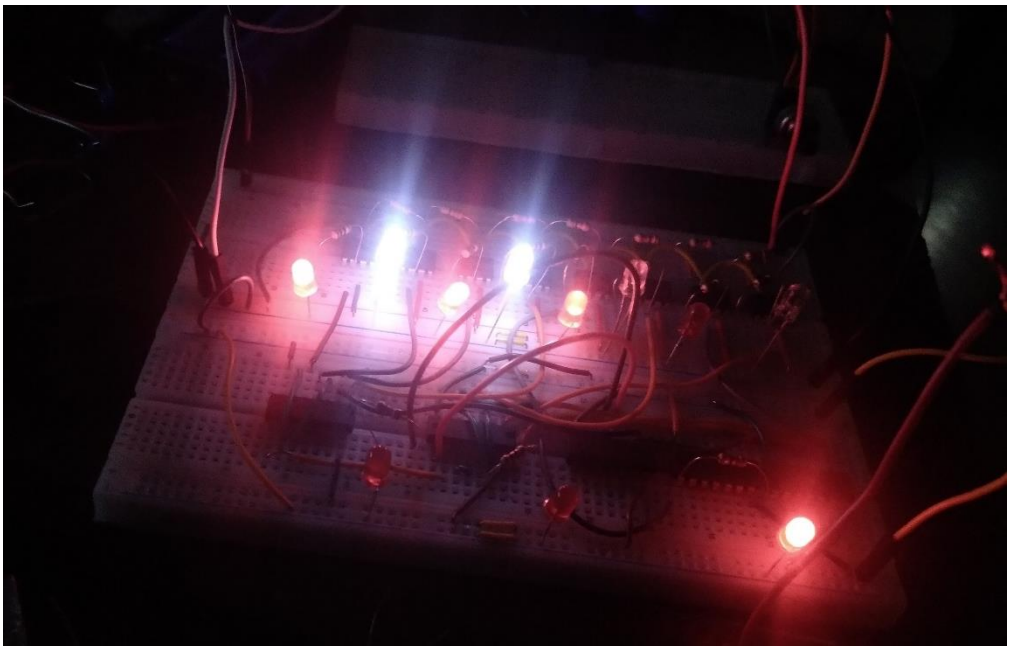
Case 3: When only three inputs are 1, and only the 2nd output LED(I2) is on:



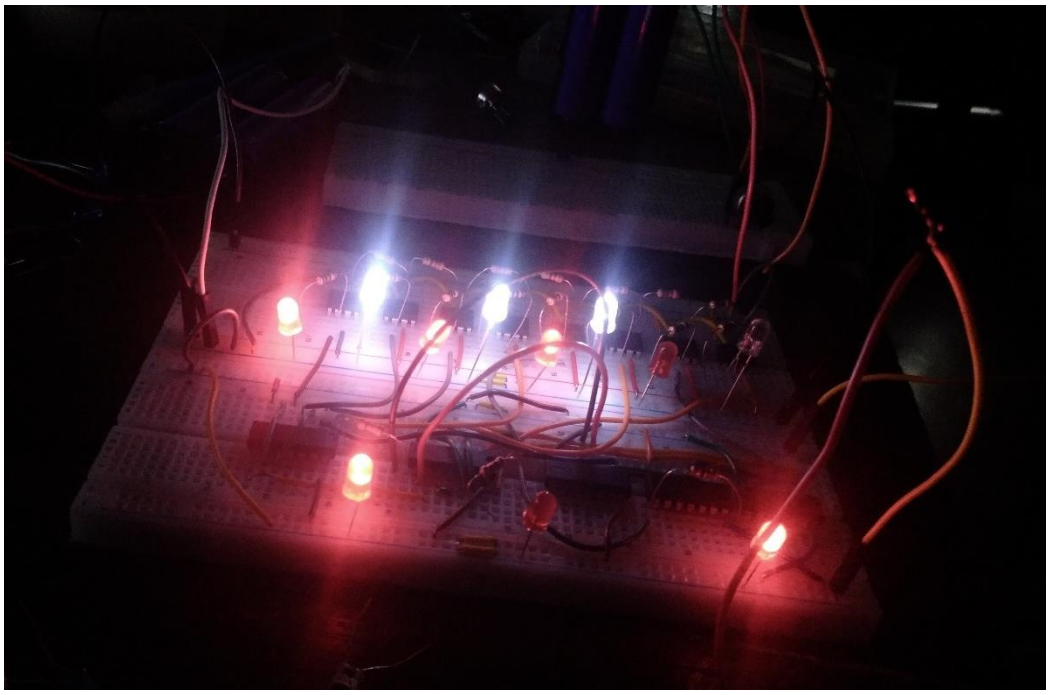
Case 4: When only four inputs are 1, and only output I2 and I3 are on:



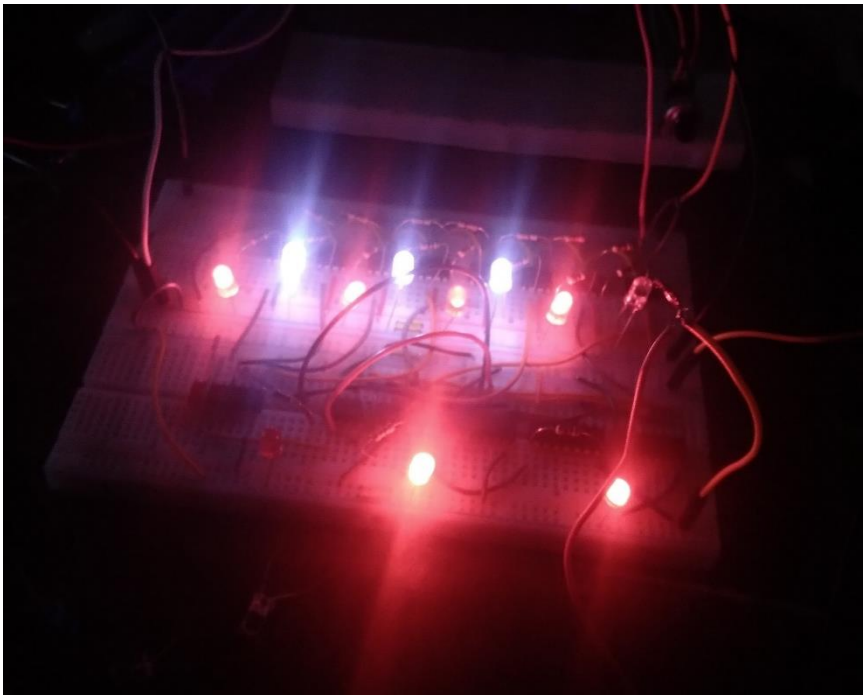
Case 5: When only 5 inputs are 1, and only output I1 is on:



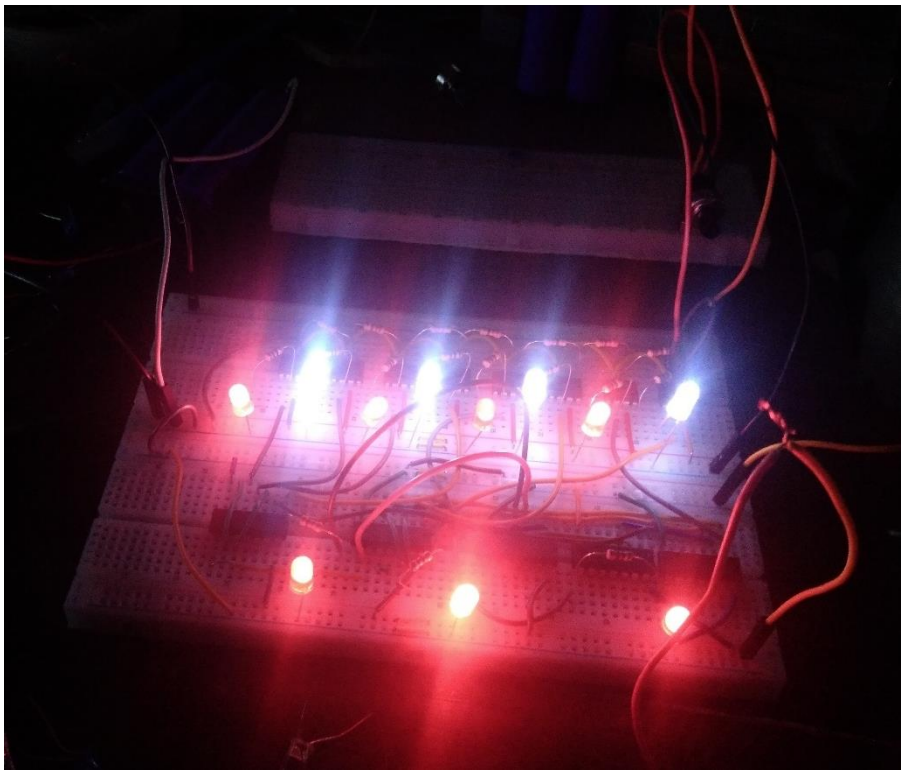
Case 6: When only 6 inputs are 1, and only output I1 and I3 is on:



Case 7: When only 7 inputs are 1, and only output I1 and I2 is on:



Case 8: When all the inputs are 1, and thus all outputs I1, I2, and I3 are also 1:



5 Design Analysis and Evaluation

5.1 Novelty

The novelty of this project lies in the circuit design and analysis of sampler, quantizer and encoder of Pulse Code Modulation (PCM) scheme. With a view to optimizing signal representation, data transmission and resilience to noise, the choice of components for sampling, quantization and encoding and development of respective circuits were conducted from scratch. Through rigorous experimentation and intuitive study, the project aims to obtain appropriate levels of performance within simplistic design.

5.2 Design Considerations (PO(c))

For a project to be both feasible and effective, design considerations are crucial. For this project, the design concerns include assessing how well the circuit can effectively sample a particular signal, quantize it to the necessary levels, and then encode it. To ensure that the output is substantially good, the findings were verified with an oscilloscope.

5.2.1 Considerations to public health and safety

The design considerations for this project prioritize public health and safety by assuring the accuracy and dependability of the circuitry. The implementation of this project upon reducing possible health hazards associated with electromagnetic radiation can significantly advance digital communication. Moreover, this project endeavors to implement robust mechanisms to ensure data security and reliability.

5.2.2 Considerations to environment

The project's environmental considerations include making sure that all implementation-related hardware and materials, including those utilized in the design and testing phases, meet environmentally friendly requirements. Efforts were made to reduce the amount of electronic waste produced throughout the testing and prototyping stages. Environmental sustainability can also be achieved by selecting hardware with low energy consumption and by designing circuits using less power. In order to reduce its carbon footprint, the project prioritizes recycling and appropriate disposal of electrical materials and components.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

At first, we designed the schematic of sampler, quantizer, and encoder in proteus. The points we considered during designing are briefed below:

- To ensure that the MOSFET works in the saturation region, we worked with maximum 1V p-p voltage.
- While biasing op-amps, biasing voltage was limited to 5V.
- Through rigorous experimentations, we found the values of resistors and capacitors.

5.3.2 Results and Analysis

The values we obtained from our project corresponded with the truth table, thus it can be said that encoding was successful.

5.4 Limitations of Tools (PO(e))

There were some limitations of some components:

- MOSFET has very difficult voltage constraints.
- There were problems working with high voltage because ICs burn.
- Also at high frequency, output is distorted.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

This project's ability to open up access to high-quality communication and multimedia technology has implications for socioeconomic and cultural issues. Through increasing the quality and efficiency of PCM systems, the project can improve communication networks and make them more dependable and accessible to a wider range of communities. Greater inclusion and connectedness may result from this, closing gaps in social services, healthcare, and education. Additionally, advancements in PCM technology can influence cultural expression through improved audio and video recording capabilities, fostering creativity and preserving cultural heritage. Overall, the project might have a good impact on cross-cultural exchanges and social connections worldwide.

5.5.2 Assessment of Health and Safety Issues

This project has a big impact on health and safety issues, especially in areas where maintaining data integrity and precise signal transmission are essential to preserving human well-being. Better Pulse Code Modulation (PCM) systems, for instance, can increase the precision and dependability of medical imaging technologies in the healthcare industry, resulting in more accurate diagnoses and better patient outcomes. Furthermore, strong PCM systems can guarantee rapid and dependable data transfer in safety-critical applications like industrial monitoring and emergency communication systems, thereby avoiding accidents and lowering dangers to human life. All things considered, the project's advances in PCM technology have the potential to significantly improve safety and health outcomes across a range of applications.

5.5.3 Assessment of Legal Issues

The impact of this project on legal issues mainly lies on ensuring compliance with regulations related to data security, privacy, and intellectual property rights. The safety of sensitive data, including medical records and personal information, requires compliance to legal frameworks. Additionally, the project may involve considerations regarding patents and copyrights for novel algorithms or developed designs.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

The team consisted of seven members, each contributing unique skills and expertise to the project:

Student Name	Contribution
Md. Nakib (2006091)	- Hardware design of sampler, simulation of sampler
Dipto Archishman Saha (2006093)	- Simulation of sampler and quantizer
Nafiul Hasan (2006094)	- Buying of components, circuit debugging
Md. Tanvirul Islam (2006095)	- Hardware design of quantizer
Md. Mostafizur Rahman (2006096)	- Hardware design of quantizer and encoder
Rony Ghosh (2006097)	- Buying of components, circuit debugging
Nahian Tasnim (2006098)	- Project planning, simulation of encoder

6.2 Mode of Team Work

To maintain strong teamwork throughout the project, the team adopted a collaborative approach. Meetings on a regular basis were used to assign tasks, review project status, and address any problems that arose. Email, messaging apps, and video chats were used as communication methods to ensure smooth collaboration, particularly when working remotely.

To ensure that responsibilities were distributed fairly depending on individual abilities and knowledge, the team established specific roles and duties for each member. Collaboration tools like Google Drive were used to share code repositories, documents, and project materials, facilitating efficient teamwork and version control.

To ensure the quality of outcomes and promote continuous improvement, the team members were encouraged to provide constructive criticism and feedback. At meetings, periodic updates were given in order to monitor project milestones and quickly handle any problems or concerns.

6.3 Diversity Statement of Team

People from different backgrounds, experiences, and viewpoints made up the team, which created a vibrant and diverse environment for collaboration. Every team member contributed unique abilities and expertise to the project, enhancing the group's overall effectiveness and creativity.

The team's diversity encouraged innovation and creativity, which made it possible to thoroughly examine the problem statement and provide workable answers. Diverse cultural backgrounds and viewpoints enriched conversations and the decision-making process, producing well-rounded conclusions and solutions.

Accepting diversity as a strength, the team utilized the abilities and ideas of its members to overcome obstacles and effectively complete the project

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

FOR IMMEDIATE RELEASE:

State-of-the-Art Project to Revolutionize Pulse Code Modulation (PCM) Systems

Dhaka, February 26, 2024 – A lucrative project, "Design of Sampler, Quantizer, and Encoder Schemes for Pulse Code Modulation," is set to explore remarkable dimensions of signal processing and communication technologies. Led by a team of students, the project aims to enhance the foundational components of PCM systems, including sampler, quantizer, and encoder designs. Through innovative approaches and rigorous experimentation, the project seeks to optimize signal quality, data transmission efficiency, and robustness to noise. Anticipated outcomes promise to drive innovation across diverse domains, propelling PCM systems into new frontiers of performance and application.

For more information, contact:

Md. Nakib

Student, Level – 3, Term – 1, Dept of EEE, BUET

Email: 2006091@eee.buet.ac.bd

8 Project Management and Cost Analysis (PO(k))

8.1Bill of Materials

Item no.	Part Name	Part Specifications	Quantity (piece)	Price (Tk)
01	AND Gate	IC 7408	1	25
03	NOT Gate	IC 7404	2	50
04	OR Gate	IC 7432	1	25
05	XOR Gate	IC 7486	2	50
06	MOSFET	IRF150	2	250
07	OP-AMP	uA741	12	240
08	555 Timer	NE555N	2	30
09	Breadboard		5	500

10	Capacitor		12	160
11	Resistor		Several	60
12	LED	Red, Blue, White	20	40
13	Li-Ion Battery	3.7V	5	300
14	Li-Ion Battery	9V	4	200
15	Connector, Jumper Wire		Several	200
	Total=			2130

9 Future Work (PO(l))

In order to extend the scope of this project in the future, the following options could be considered:

1. *Advanced Sampling Techniques:* Exploring novel sampling methods such as compressive sensing or adaptive sampling to optimize quality and efficiency in capturing analog signals.
2. *Quantization Optimization:* Investigating advanced quantization techniques such as non-uniform quantization or adaptive quantization algorithms to achieve higher signal-to-noise ratios.
3. *Hardware Implementation:* Examining hardware-efficient implementations of sampler, quantizer, and encoder components.
4. *Application-specific Modification:* Customizing PCM systems to particular application domains, such as Internet of Things (IoT) devices, medical imaging, or telecommunications, taking into account particular requirements and limits.

10 References

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